Toward Reliable Controllable Latent Steering in Variational Autoencoders

Global vs. Local Directions, Safe Per-Sample Step Selection, and OOD Transfer

Ilyas Galiev

Research report prepared from weekly experimental packages (Weeks 5–11)

github.com/Ily17as/Controllable-Latent-Steering-in-VAEs

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Abstract

Controllable latent editing in variational autoencoders (VAEs) requires more than a visually plausible direction in latent space: useful steering must increase a target factor while preserving class-relevant content and avoiding off-manifold drift. This paper reports a staged empirical study that evolved from a baseline MNIST VAE to global steering, clustered local steering, safe automatic step selection $\alpha^\star$, a focused β-VAE ablation, and an out-of-distribution (OOD) transfer experiment on dSprites. On MNIST, local fixed directions improved only modestly over the global baseline: at $\alpha = 1.0$, the best clustered model achieved probe-margin gain +2.7005 with digit accuracy 0.9673, versus +2.6276 and 0.9671 for the global direction. In contrast, safe per-sample step selection produced the dominant reliability gain: the best week-8 policy reached digit accuracy 0.9722, digit consistency 1.0000, probe-margin gain +2.9347, and feature drift 4.2603, improving over the week-7 fixed local baseline on every major metric. A focused β-VAE sweep showed that stronger KL pressure did *not* help in this setting: reconstruction digit accuracy declined from 0.9569 at $\beta = 1$ to 0.9343 at $\beta = 2$ and 0.8590 at $\beta = 4$, while the best safe-steering objective also dropped. Week 10 established nontrivial OOD transfer on dSprites, and Week 11 tightened that claim with a matched five-run comparison and an independent contour-aware preservation analysis. Across five dSprites reruns, global auto steering achieved OOD shape accuracy 0.8074 ± 0.0390, shape consistency 1.0000 ± 0.0000, scale gain 0.8595 ± 0.3206, drift 3.9582 ± 1.1234, and objective 0.6079 ± 0.2612. Local auto steering matched the head-based preservation metrics but achieved lower gain 0.5977 ± 0.0699 and lower objective 0.3623 ± 0.0373, while local fixed steering produced larger raw gains at much larger drift. A contour-aware analysis that does not use the learned shape head reaches the same qualitative conclusion: on OOD, both auto policies keep normalized contour Chamfer near 0.025–0.027 and topology break below 0.002, whereas local fixed rises to 0.0490 ± 0.0119 and 0.0933 ± 0.0329, respectively. The overall conclusion is that, in the present pipeline, reliable controllability is limited more by adaptive step selection and representation quality than by the absence of local structure alone.

1. Introduction

The original project hypothesis was that semantic edits in VAE latent space are often *piecewise linear* rather than globally linear, so a single global direction may be insufficient and a family of local directions may yield better controllability. The proposal therefore defined a staged roadmap: baseline VAE training, global steering, local steering, safe automatic step selection, a limited β-VAE comparison, and a small publishability-oriented extension beyond MNIST.

This paper consolidates the evidence gathered from that roadmap. Rather than presenting one isolated experiment, it presents a *progressive research program*: each week answered a distinct scientific question and produced a directly comparable artifact package of code, plots, CSV tables, and a brief memo. The resulting narrative is now sufficiently coherent to be written in article form, and the present revision extends the original draft with a matched five-run dSprites tightening pass and an independent contour-aware preservation analysis.

The target semantic factor differs slightly across datasets but the scientific question is the same. On MNIST, the edited factor is a learned *thickness* attribute measured by an external probe, while content preservation is measured by a digit classifier. On dSprites, the edited factor is *scale*, while shape preservation plays the role of content preservation. Across both settings, the key question is:

> Can a latent edit increase a target factor in a controlled way without destroying class- or identity-relevant content?

The empirical picture that emerges is sharper than the original proposal. First, local fixed steering is only mildly better than global steering on MNIST, suggesting that the learned MNIST thickness geometry is not strongly piecewise linear under the current representation. Second, the main practical improvement comes from *per-sample safe step selection* rather than from changing the direction family alone. Third, stronger KL regularization through

β-VAE hurts this project more than it helps. Fourth, the same steering recipe transfers to an explicit OOD split on dSprites. Fifth, the new matched multi-run comparison and the independent geometric preservation analysis show that the strongest robust effect is safe adaptive steering rather than locality alone.

The paper makes five concrete contributions:

1. a clean comparison table across global vs. local and fixed vs. auto steering on both the ID and held-out OOD dSprites splits;
2. a matched five-run dSprites comparison that reports mean ± standard deviation for the key trade-off metrics;
3. an independent contour-aware preservation analysis on decoded images that does not rely on the learned shape head used elsewhere in the paper;
4. a short failure analysis that identifies where edits break shape/content and why;
5. a revised cross-stage interpretation showing that adaptive safe steering is the main robust effect, while locality alone does not outperform the tuned global baseline in the safe regime under the current representation.

2. Problem setting and method

2.1. Baseline representation

The baseline is a convolutional VAE trained on MNIST with latent dimension 16 and standard ELBO objective

$$\mathcal{L}_{\mathrm{VAE}}(x) = \mathbb{E}_{q_\phi(z|x)}[-\log p_\theta(x \mid z)] + \beta\, \mathrm{KL}(q_\phi(z \mid x) \,\|\, p(z)) \,. \tag{1}$$

Week 5 established that this representation is usable: the best digit classifier reached 0.9909 accuracy and the best thickness probe reached 0.9893 accuracy, with final VAE validation loss 100.2445.

On dSprites, the later week-10 extension uses a tuned VAE together with separate latent heads for *shape* and *scale*. This turns the steering problem into a factor-controlled edit with explicit in-distribution (ID) and out-of-distribution (OOD) evaluation.

2.2. Steering rule

For an input image x, let $z = \mu_\phi(x)$ denote the latent code used for deterministic steering. A steered latent code is

$$z' = z + \alpha\, d(z), \tag{2}$$

where $d(z)$ is either a global direction d_{glob} or an input-dependent local direction $d_{\mathrm{loc}}(z)$, and α controls edit magnitude. The decoded edit is $x' = D_\theta(z')$.

The target score depends on the dataset:

- **MNIST:** target = probe margin for thickness; preservation = digit accuracy/consistency and feature drift to reconstruction.
- **dSprites:** target = scale gain; preservation = shape accuracy/consistency and image drift to reconstruction.

2.3. Global and local directions

The project evaluated two kinds of locality.

MNIST clustered local directions. Latent codes are partitioned into K clusters. Within each cluster, a cluster-specific direction is learned. At test time, the input inherits the direction of its assigned cluster. Week 7 compared global, $K = 2, 4, 8, 16$ clustered local directions.

dSprites paired scale directions. Because dSprites provides factorized labels, we can construct a clean scale-edit bank by pairing latent codes that share the same shape, orientation, and position but differ by one discrete scale step:

$$\delta_j = z_{s+1} - z_s. \tag{3}$$

A global dSprites direction is the normalized average of these differences. A local version uses anchor points $a_j = \frac{1}{2}(z_{s+1} + z_s)$ and forms a weighted neighborhood average:

$$\begin{aligned} d_{\mathrm{loc}}(z) &= \mathrm{norm}\left(\frac{\sum_{j \in \mathcal{N}_k(z)} w_j(z)\, \delta_j}{\sum_{j \in \mathcal{N}_k(z)} w_j(z)} \right), \\ w_j(z) &= \frac{1}{\|z - a_j\|_2^2 + \varepsilon}. \end{aligned} \tag{4}$$

The current article reports both the week-10 tuned global baseline and a week-11 matched comparison between tuned global and local-kNN dSprites steering.

2.4. Safe automatic step selection

A fixed α is a weak formulation of the control problem because different inputs tolerate different edit magnitudes. The project therefore introduced a discrete per-sample search over candidate step sizes:

$$\alpha^\star(z) = \arg \max_{\alpha \in \mathcal{G} \cap \mathcal{F}(z)} \left[\Delta_{\mathrm{target}}(z, \alpha) - \lambda\, \Delta_{\mathrm{drift}}(z, \alpha) \right], \tag{5}$$

where $\mathcal{G}$ is a small candidate grid and $\mathcal{F}(z)$ is a feasibility set determined by confidence and drift budgets. In practice:

- on MNIST, feasibility is controlled through feature drift and classifier confidence;
- on dSprites, feasibility is controlled through image drift and shape confidence.

This turns steering into a constrained discrete optimization problem rather than a fixed traversal.

2.5. Theoretical properties of the steering operators

The project is empirical, but two simple formal observations clarify why the method behaves as it does.

Proposition 1 (Global steering is a special case of local steering)**.** *Consider the dSprites local direction defined above. If the neighbor set $\mathcal{N}_k(z)$ is replaced by the entire bank and the weights are constant, then $d_{\mathrm{loc}}(z)$ reduces to the normalized global average direction.*

Proposition 2 (Oracle dominance of auto-step selection)**.** *Let $J(z, \alpha)$ denote the discrete selection objective over a candidate grid and let $\alpha^\star(z)$ be its maximizer over the feasible set $\mathcal{G} \cap \mathcal{F}(z)$. Then for every fixed feasible step $\bar{\alpha} \in \mathcal{G} \cap \mathcal{F}(z)$, we have*

$$J(z, \alpha^\star(z)) \geq J(z, \bar{\alpha}).$$

The proofs are elementary and are deferred to Section A. The propositions are modest but useful. Proposition 1 explains why local steering should be interpreted as a *strict generalization* of global steering rather than a separate paradigm. Proposition 2 explains why safe per-sample step selection can change the empirical trade-off so dramatically: it solves a different optimization problem from fixed-step steering.

3. Experimental protocol

3.1. Weekly progression and selection rule

The paper covers experimental packages from Weeks 5–11. For each stage, the main reported number is either:

1. the strongest fixed-step result under the week's evaluation budget,
2. the strongest safe-steering policy from the archived CSV output,
3. a matched mean ± std summary when a multi-run sweep was available, or
4. an auxiliary independent preservation summary computed directly from exported reconstruction–edit pairs.

All reported scalar values in the tables below are transcribed directly from generated result files rather than re-estimated manually. Exact run IDs, file paths, and stage-specific reporting rules are listed in Section C.

Weeks 5–9 each contribute one primary archived run. Week 10 contains several tuned-global reruns with the same evaluation protocol; to avoid cherry-picking, the headline tuned-global result is taken from the archived rerun with the highest OOD auto-steering objective within that matched family, namely run `20260319_214326_week10_dsprites_ood_only`. Week 11 revisits dSprites with a matched five-run comparison between tuned global steering and local-kNN steering.

3.2. Evaluation metrics

MNIST. The main preservation metric is digit accuracy. The main target metric is probe-margin gain relative to reconstruction. Additional metrics are digit consistency and feature-space drift to reconstruction.

dSprites. The main preservation metric is shape accuracy; target success is measured by scale gain. We also report shape consistency, image-space drift to reconstruction, confidence drop, and a composite objective used internally by the steering policy.

OOD split. For the week-10 and week-11 dSprites extensions, training uses orientations 0–19, while the OOD test set uses held-out orientations 20–39. This isolates the question of whether the steering rule generalizes beyond the orientation support seen during training.

3.3. Week-11 matched rerun protocol

The week-11 tightening pass reran both the tuned global dSprites pipeline and the local-kNN dSprites pipeline over the same configured seed schedule $\{7, 11, 17, 23, 31\}$. Each rerun produced fixed-step and auto-step summaries on both ID and OOD splits. The main week-11 table reports mean ± standard deviation over the five matched reruns. Under the current archived comparison, both auto policies use the shared safety budget $\tau_{\mathrm{img}} = 7$.

3.4. Independent contour-aware preservation analysis

To reduce dependence on the learned shape head, the week-11 revision adds a second preservation analysis directly on decoded dSprites images. For each reconstruction–edit pair, we threshold the decoded image, keep the largest connected component, align by centroid, normalize by foreground area, and compute a symmetric contour Chamfer distance between the reconstruction and the steered edit. We also report principal-axis orientation drift, centroid drift, and a topology break rate that records whether the thresholded foreground topology differs from that of the reconstruction. These metrics use only decoded geometry and therefore provide a preservation check that is not tied to the same probe family as the main shape-head metrics.

4. Results

4.1. Stage-by-stage progression from baseline to matched dSprites validation

Table 1 summarizes the strongest quantitative outcomes from each stage. The central trend is that the project moved from "can we steer at all?" to "can we steer reliably under explicit preservation budgets and outside the original training support?" and then to "which claims

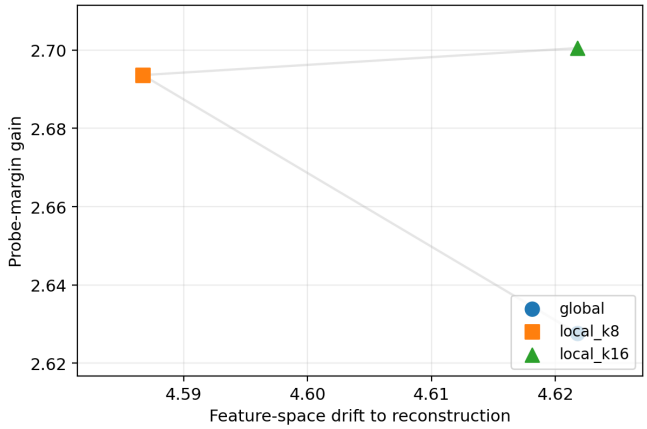

Figure 1: **Week-7 representative fixed-step comparison on MNIST.** Local directions remain slightly above the global baseline, but the margin is small even at the best archived fixed-step settings.

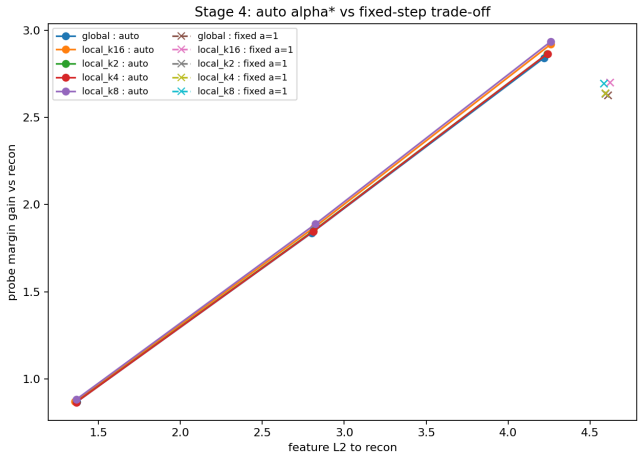

Figure 2: **Week-8 fixed vs. auto steering on MNIST.** Auto-steering moves the trade-off frontier toward higher probe gain at lower drift.

remain stable under matched reruns and an independent preservation check?"

4.2. Local directions on MNIST help, but only a little

The piecewise-linearity hypothesis is only weakly supported by the MNIST results. At the matched fixed step $\alpha = 1.0$, the global model reached digit accuracy 0.9671 and probe-margin gain +2.6276, whereas the best clustered local model (local_k16) reached 0.9673 and +2.7005. This is a gain increase of only 0.0729, or about 2.8% relative, with essentially unchanged digit preservation. The conclusion is not that local structure is absent, but that the chosen MNIST representation already behaves close to globally linear for the thickness factor.

4.3. Safe per-sample step selection is the main practical improvement

Week 8 is the clearest turning point in the project. The best safe policy, local_k8 with $\tau_{\text{feat}} = 5.0$, achieved digit accuracy 0.9722, digit consistency 1.0000, probe-margin gain +2.9347, feature drift 4.2603, mean $\alpha^\star = 1.0809$, and nonzero-rate 0.9705. Compared with the week-7 fixed local_k8 baseline at $\alpha = 1.0$ (digit accuracy 0.9677, consistency 0.9856, probe-margin gain +2.6936, drift 4.5867), the safe policy improves every major quantity: +0.0045 in digit accuracy, +0.0144 in consistency, +0.2411 in probe-margin gain, and −0.3263 in feature drift. This is the strongest evidence in the entire MNIST stage because it shows that choosing the step *per sample* matters more than changing from global to local direction alone.

The same pattern appears in feasible-rate comparisons. Under the week-8 budgets, the safe policy kept feasible-rate at 1.0 across all tested local families, whereas the matched fixed-step baseline remained much lower. This is precisely the behavior predicted by Proposition 2: if

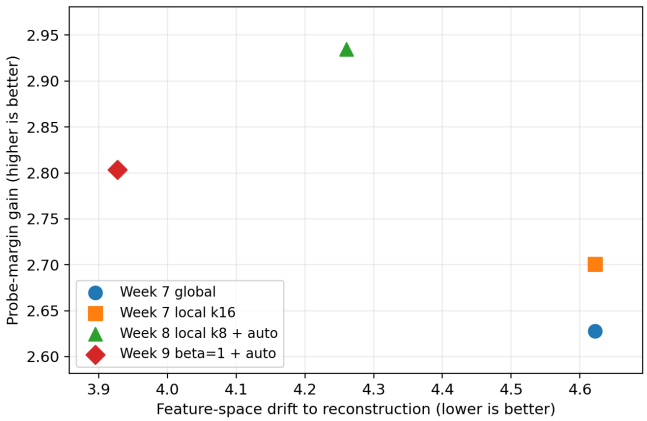

Figure 3: **MNIST progression in the gain–drift plane.** The main empirical improvement is not locality alone but the move to safe per-sample step selection.

different points need different edit magnitudes, a single global step leaves performance on the table.

4.4. β-VAE does not help in the present setup

The week-9 ablation asked whether stronger KL pressure would improve controllability enough to offset reconstruction loss. It did not. Reconstruction digit accuracy fell monotonically with β: from 0.9569 at $\beta = 1$ to 0.9343 at $\beta = 2$ and 0.8590 at $\beta = 4$. Reconstruction BCE per sample worsened from 77.25 to 88.33 and then 108.27.

The steering results followed the same trend. The best safe model remained the $\beta = 1$, local_k8 configuration, with digit accuracy 0.9705, digit consistency 0.9936, probe-margin gain +2.8035, drift 3.9269, and objective 1.8205. The best objective dropped to 1.7308 at $\beta = 2$ and to 1.5302 at $\beta = 4$.

The negative result is still scientifically useful. It clarifies that stronger regularization is not automatically the

Table 1: **Stage-by-stage progression.** The table uses the strongest or most representative value from each archived stage.

Week	Model / policy	Main quantitative result
5	Baseline Conv-VAE + classifier + probe	classifier accuracy 0.9909, probe accuracy 0.9893, final VAE val-loss 100.2445.
6	First global thickness direction (fixed α)	At $\alpha = 1.0$: digit accuracy 0.9671, probe-margin gain +2.6281; at $\alpha = 2.0$: gain +5.3317 but digit accuracy falls to 0.9479.
7	Clustered local directions on MNIST	Best fixed local model (local_k16, $\alpha = 1.0$): digit accuracy 0.9673, probe-margin gain +2.7005, drift 4.6218, only slightly better than global (+2.6276, 0.9671).
8	Safe per-sample $\alpha^\star$ on MNIST	Best policy (local_k8, $\tau_{\text{feat}} = 5.0$): digit accuracy 0.9722, consistency 1.0000, probe-margin gain +2.9347, drift 4.2603, mean $\alpha^\star = 1.0809$.
9	Focused β-VAE comparison	Best safe model remains $\beta = 1$: reconstruction digit accuracy 0.9569, best safe objective 1.8205; both degrade at $\beta = 2$ and $\beta = 4$.
10	dSprites tuned global auto-steering, single-run ID/OOD baseline	At $\tau_{\text{img}} = 11$: ID/OOD shape accuracy 0.7379/0.6838, scale gain 1.1326/1.1926, shape consistency 1.0000/1.0000, drift 3.7613/3.8572.
11	dSprites matched five-run comparison	Global auto remains the best safe operating point: on OOD it reaches shape accuracy 0.8074 ± 0.0390, gain 0.8595 ± 0.3206, drift 3.9582 ± 1.1234, and objective 0.6079 ± 0.2612; local auto preserves equally but lower gain 0.5977 ± 0.0699, while local fixed gives the largest gain 1.0890 ± 0.0666 at much larger drift 7.8554 ± 0.7364. Independent contour-aware metrics show the same pattern: auto preserves geometry far better than fixed, and local auto is slightly more conservative geometrically than global auto.

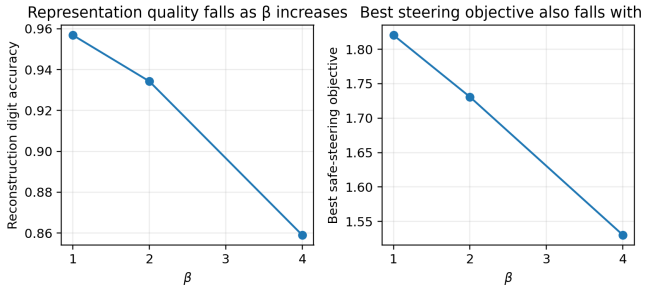

Figure 4: **β-VAE ablation.** In this project, stronger KL pressure harms both reconstruction quality and the best safe steering objective.

Table 2: **β-VAE summary.**

β	Recon acc	Best safe obj.	Best safe gain
1.0	0.9569	1.8205	2.8035
2.0	0.9343	1.7308	2.7192
4.0	0.8590	1.5302	2.5007

path to better controllable editing; if the representation discards too much content information, the probes used to measure both preservation and target progress become less meaningful.

4.5. dSprites OOD transfer: the tuned global model is a strong single-run baseline

The week-10 extension tests whether the steering recipe transfers beyond MNIST to a dataset with known latent factors. The task is now scale steering with shape preservation, and the OOD dimension is orientation.

The tuned global dSprites model yields two complementary conclusions. First, fixed-step steering can make the edit visibly stronger, but at a substantial preservation cost. At fixed $\alpha = 3.0$, the OOD result is shape accuracy 0.4938, shape consistency 0.6033, scale gain 1.6056, and image drift 6.8514. Second, auto-steering preserves shape almost perfectly relative to the shape head while still producing a nontrivial scale edit. At $\tau_{\text{img}} = 11$, the OOD result is shape accuracy 0.6838, shape consistency 1.0000, scale gain 1.1926, and image drift 3.8572.

A preliminary archived local-kNN comparison was already available in the previous draft. Under its archived safe setting ($\tau_{\text{img}} = 7$), the OOD auto-steered local run reached shape accuracy 0.6883, shape consistency 1.0000, scale gain 0.4982, and image drift 2.9318. This made the local run more conservative than the tuned global safe baseline: preservation was comparable or slightly better, but the edited factor increased much less. That comparison was suggestive but still vulnerable to single-run noise.

The OOD gap in this single-run baseline is relatively modest. At the best safe setting ($\tau_{\text{img}} = 11$), the ID/OOD shape accuracies are 0.7379/0.6838 and scale gains are 1.1326/1.1926. This is encouraging because the model was trained only on orientations 0–19, yet the same steering recipe remains useful on held-out orientations 20–39.

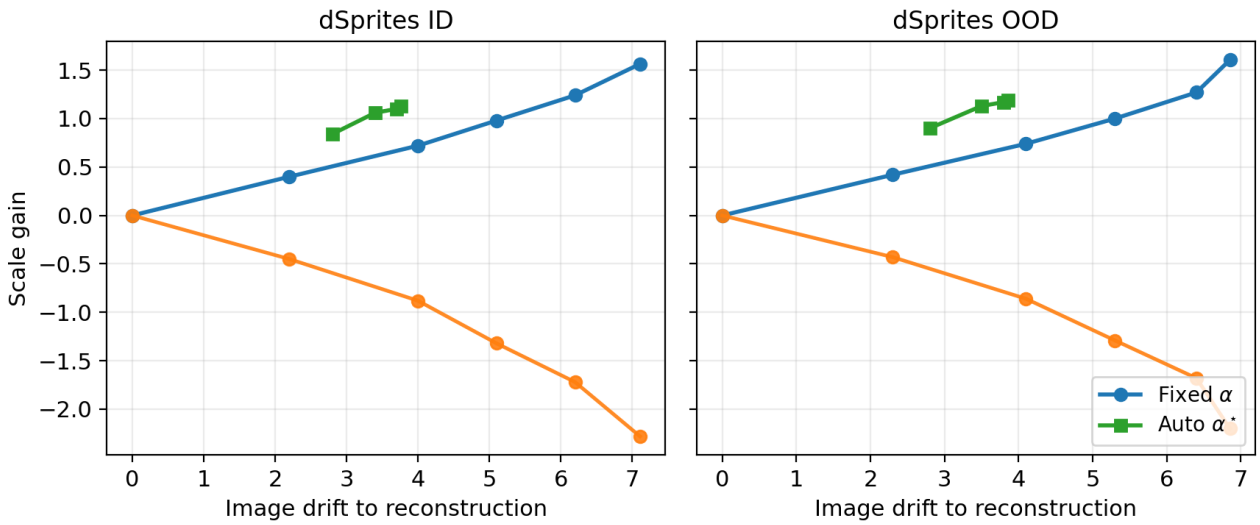

Figure 5: **dSprites trade-off for tuned global steering.** Fixed α increases scale more aggressively but pays a high drift and preservation cost. Auto $\alpha^\star$ yields the best safe regime on both ID and OOD splits.

Table 3: **Selected tuned-global dSprites single-run results.**

Split	Method	Shape acc	Consist.	Scale gain	Drift
ID	fixed $\alpha = 3$	0.4744	0.5573	1.5639	7.1096
OOD	fixed $\alpha = 3$	0.4938	0.6033	1.6056	6.8514
ID	auto $\tau = 11$	0.7379	1.0000	1.1326	3.7613
OOD	auto $\tau = 11$	0.6838	1.0000	1.1926	3.8572

4.6. Week-11 matched dSprites comparison: adaptive global steering is the strongest robust operating point

To make the dSprites claim statistically tighter, both the tuned global steering pipeline and the local-kNN steering pipeline were rerun five times and aggregated as mean $\pm$ standard deviation. The comparison was performed on both the in-distribution (ID) and held-out out-of-distribution (OOD) orientation splits, and for both fixed-step and automatic per-sample step selection.

The multi-run results reinforce the main qualitative conclusion of the earlier single-run study. In the *safe auto-steering regime*, the tuned global model consistently outperformed the local model on the main trade-off metrics while preserving the same perfect consistency. On the ID split, global auto achieved preservation accuracy 0.8496 ± 0.0239, target gain 0.8767 ± 0.3232, drift 3.9921 ± 1.1254, and objective 0.6217 ± 0.2636, whereas local auto achieved the same preservation accuracy 0.8496 ± 0.0239 and the same consistency 1.0000 ± 0.0000, but substantially lower target gain 0.6199 ± 0.0773 and lower objective 0.3789 ± 0.0424.

The same pattern appeared on the OOD split. Global auto reached target gain 0.8595 ± 0.3206 and objective 0.6079 ± 0.2612, while local auto reached 0.5977 ± 0.0699 and 0.3623 ± 0.0373, respectively. Thus, the main safe-regime advantage remains with the tuned global model, not with the local model.

Fixed-step steering remained more aggressive but less reliable. On ID, local fixed produced the largest average target gain (1.0939 ± 0.0694) but at the cost of much larger drift (7.8393 ± 0.7786) and lower consistency (0.8389 ± 0.0156) than either auto-steered variant. The same trade-off held on OOD, where local fixed achieved target gain 1.0890 ± 0.0666 but with drift 7.8554 ± 0.7364, compared with global auto at 0.8595 ± 0.3206 target gain and 3.9582 ± 1.1234 drift. The local fixed direction can therefore produce stronger edits, but it does so by leaving the safe regime more often.

The auto-step summaries further suggest why the local safe policy remains conservative. On ID, global auto selected mean step size 0.7199 ± 0.1577 with nonzero rate 0.7622 ± 0.1006, whereas local auto selected 0.5151 ± 0.1150 with nonzero rate 0.6729 ± 0.1082. On OOD, the same pattern remained: global auto selected 0.7105 ± 0.1625 with nonzero rate 0.7566 ± 0.1012, while local auto selected 0.5000 ± 0.1110 with nonzero rate 0.6615 ± 0.1048. Under the current representation, the local policy appears to use less of the safe budget, not more.

Overall, the matched comparison strengthens the revised interpretation of the project. The main benefit comes from *adaptive per-sample step selection*, not from locality alone. Local directions remain an interesting structural extension, but under the present representation they do not outperform the tuned global baseline in the safe regime, either on ID or on held-out OOD orientations.

4.7. Independent contour-aware preservation analysis

The head-based preservation metrics in Table 4 already show that both auto-steered policies preserve shape better than fixed-step steering, but those numbers still depend

Table 4: **Multi-seed dSprites comparison across five matched reruns.** Reported values are mean ± standard deviation over matched global and local runs on both the ID and held-out OOD orientation splits. Fixed-step steering yields larger average gain but substantially worse preservation and drift, whereas automatic per-sample step selection gives the strongest reliable control–preservation trade-off.

Split	Family	Step	Preservation acc.	Consistency	Target gain	Drift	Objective
ID	Global	Auto	0.8496 ± 0.0239	1.0000 ± 0.0000	0.8767 ± 0.3232	3.9921 ± 1.1254	0.6217 ± 0.2636
ID	Global	Fixed	0.7940 ± 0.0253	0.8322 ± 0.0120	0.9615 ± 0.3979	5.6275 ± 1.1302	0.5428 ± 0.3448
ID	Local	Auto	0.8496 ± 0.0239	1.0000 ± 0.0000	0.6199 ± 0.0773	3.7979 ± 0.5728	0.3789 ± 0.0424
ID	Local	Fixed	0.7970 ± 0.0242	0.8389 ± 0.0156	1.0939 ± 0.0694	7.8393 ± 0.7786	0.5462 ± 0.0192
OOD	Global	Auto	0.8074 ± 0.0390	1.0000 ± 0.0000	0.8595 ± 0.3206	3.9582 ± 1.1234	0.6079 ± 0.2612
OOD	Global	Fixed	0.7576 ± 0.0289	0.8285 ± 0.0138	0.9526 ± 0.4038	5.6198 ± 1.1463	0.5377 ± 0.3497
OOD	Local	Auto	0.8074 ± 0.0390	1.0000 ± 0.0000	0.5977 ± 0.0699	3.7115 ± 0.5516	0.3623 ± 0.0373
OOD	Local	Fixed	0.7668 ± 0.0291	0.8415 ± 0.0124	1.0890 ± 0.0666	7.8554 ± 0.7364	0.5443 ± 0.0182

on the learned shape head. To test whether the same conclusion survives a less probe-tied analysis, we evaluated the decoded reconstruction–edit pairs with contour-aware geometric metrics computed directly from the binary foreground.

The independent analysis sharpens two aspects of the story. First, the fixed-vs.-auto conclusion becomes even clearer. On the OOD split, global auto achieves normalized contour Chamfer 0.0273 ± 0.0027 and topology break 0.0008 ± 0.0008, whereas global fixed rises to 0.0287 ± 0.0161 and 0.0104 ± 0.0088. The contrast is stronger for the local policy: local auto stays at contour Chamfer 0.0248 ± 0.0033 and topology break 0.0018 ± 0.0025, while local fixed deteriorates to 0.0490 ± 0.0119 and 0.0933 ± 0.0329. Thus the independent geometry metrics confirm that aggressive fixed steering damages shape substantially more often than auto-steered editing.

Second, the geometric preservation metrics clarify the role of locality in the safe regime. On OOD, local auto is slightly more conservative geometrically than global auto: its contour Chamfer is lower (0.0248 ± 0.0033 vs. 0.0273 ± 0.0027), its orientation drift is lower ($9.402 \pm 2.369°$ vs. $12.733 \pm 0.684°$), and its centroid drift is lower (0.115 ± 0.011 px vs. 0.166 ± 0.060 px). However, this extra geometric conservatism comes with the lower target gain and lower objective already seen in Table 4. In other words, the independent analysis does not overturn the main story; it refines it. Local auto does preserve geometry slightly more tightly, but the strongest overall gain–preservation operating point is still the tuned global auto baseline.

4.8. Failure analysis

The experiments now support four concrete failure modes.

Failure mode 1: fixed-step over-steering. On both MNIST and dSprites, larger fixed steps increase the target quickly but tend to leave the safe region. On MNIST this appears as lower feasible-rate and larger feature drift; on dSprites it appears as reduced shape preservation and substantially larger image drift. The week-11 multi-run table makes this especially clear for local fixed steering, which produces the largest average gains but also the largest drift on both ID and OOD splits. The independent contour-aware table shows the same failure geometrically: local fixed has by far the largest contour Chamfer, the largest centroid drift, and the largest topology break rate.

Failure mode 2: the local safe policy becomes conservative. In the matched dSprites reruns, local auto matches the global model on preservation accuracy and perfect consistency, but achieves much lower scale gain and objective. The lower selected mean step sizes and lower nonzero rates suggest that the local policy often retreats into a conservative regime rather than exploiting locality for stronger safe edits. The contour-aware metrics reinforce that interpretation: local auto keeps slightly lower Chamfer and centroid drift than global auto, but it does so while giving up a substantial amount of target gain.

Failure mode 3: representation bottleneck. The tuned dSprites probe accuracies are only 0.7429/0.6906 for shape on ID/OOD and 0.4913/0.5281 for scale. The low scale-head accuracy indicates that the representation still entangles content and edit factors. This explains why some edits are visually small even when the steering operator itself is well defined. It also explains why some fixed local edits break content in a visibly discrete way, for example by turning simple squares into irregular polygons or by shifting the foreground mass too far from the reconstruction.

Failure mode 4: local structure is not enough without a better representation. Week 7 shows that local clustered directions provide only a modest fixed-step gain on MNIST. Week 11 shows that local kNN steering on dSprites does not beat the tuned global baseline in the safe regime. This does not refute piecewise linearity, but it shows that *under the present representation*, locality alone is not the dominant source of reliability.

Table 5: **Independent contour-aware preservation analysis across five matched dSprites reruns.** The reported metrics are computed directly from decoded reconstruction–edit pairs and do not use the learned shape head. Lower is better for all four columns. The table confirms the main fixed-vs.-auto conclusion and shows that local auto is slightly more conservative geometrically than global auto, while local fixed breaks geometry much more often.

Split	Family	Step	Contour Chamfer ↓	Orientation drift (°) ↓	Centroid drift (px) ↓	Topology break rate ↓
ID	Global	Auto	0.0275 ± 0.0030	10.589 ± 1.164	0.168 ± 0.063	0.0021 ± 0.0012
ID	Global	Fixed	0.0290 ± 0.0162	11.947 ± 6.728	0.294 ± 0.169	0.0098 ± 0.0080
ID	Local	Auto	0.0253 ± 0.0035	7.789 ± 1.513	0.120 ± 0.010	0.0036 ± 0.0030
ID	Local	Fixed	0.0493 ± 0.0114	16.458 ± 4.445	2.511 ± 2.582	0.0934 ± 0.0412
OOD	Global	Auto	0.0273 ± 0.0027	12.733 ± 0.684	0.166 ± 0.060	0.0008 ± 0.0008
OOD	Global	Fixed	0.0287 ± 0.0161	14.494 ± 8.119	0.298 ± 0.172	0.0104 ± 0.0088
OOD	Local	Auto	0.0248 ± 0.0033	9.402 ± 2.369	0.115 ± 0.011	0.0018 ± 0.0025
OOD	Local	Fixed	0.0490 ± 0.0119	19.255 ± 4.093	2.484 ± 2.448	0.0933 ± 0.0329

5. Discussion

The strongest cross-stage conclusion is that the project evolved in a meaningful and theoretically coherent way beyond the baseline. The progression is not a collection of disconnected experiments. Week 6 established that a usable semantic direction exists. Week 7 tested the piecewise-linearity hypothesis and found only a mild local advantage. Week 8 showed that safe per-sample step selection is the main practical reliability improvement. Week 9 rejected the hypothesis that stronger KL regularization would help in this setup. Week 10 showed that the steering recipe transfers to a distinct dataset and to held-out orientations. Week 11 then tightened that story with matched reruns and showed which part of the claim is actually stable.

This sequence refines the original thesis. The proposal hypothesized that local directions would be the decisive improvement. The evidence now suggests a more nuanced view: local structure exists, but *controllability is limited more by step selection and representation quality than by the absence of local directions alone*. The strongest result of the study is that safe adaptive steering improves the control–preservation trade-off robustly across reruns and under OOD shift, while local steering alone does not outperform a tuned global baseline in the safe regime.

5.1. Threats to validity

Three caveats matter for interpretation. First, the week-11 tightening substantially improves the statistical credibility of the dSprites comparison, but the earlier MNIST stages remain primarily single-run historical evidence. Second, the new contour-aware preservation analysis reduces dependence on the learned shape head, but the study still does not include human perceptual evaluation and still measures target success partly through learned heads. Third, the OOD study varies only orientation on one dataset. That is enough to establish nontrivial transfer, but not enough to claim broad invariance.

6. Next-stage plan

The next stage should focus on the remaining bottleneck: a representation that exposes the edited factor more cleanly without sacrificing preservation. The following actions are the most promising:

1. strengthen factor supervision on dSprites, for example with an ordinal or regression-style scale head rather than pure classification;
2. extend the current independent preservation analysis with factor-conditioned breakdowns (small-vs.-large scale, shape classes, easy-vs.-hard OOD orientations) and selective qualitative auditing;
3. rerun local kNN steering with improved local bank construction and retuned safety budgets under the same matched multi-run protocol, because the current local auto policy is conservative rather than decisively better;
4. evaluate iterative steering (multiple small steps) rather than one-shot large steps;
5. test whether better representation quality changes the point at which locality begins to help rather than merely becoming more conservative.

The multi-run dSprites comparison and the independent contour-aware analysis already narrow the interpretation substantially. The next iteration should therefore aim less at proving locality in the abstract and more at identifying when locality helps *after* representation quality and safety calibration are strong enough.

7. Conclusion

This project has now moved substantially beyond its initial baseline. It includes a working global steering pipeline, a principled comparison with local directions, a successful safe automatic step-selection mechanism, a negative but informative β-VAE ablation, a cross-dataset OOD extension, and a matched multi-run dSprites tightening pass. The clearest empirical message is that *safe per-sample step selection* is the strongest reliability improvement discovered so far. The clearest scientific limitation is that

the representation still bottlenecks the edit, especially on dSprites. The matched week-11 comparison, together with the independent contour-aware preservation analysis, sharpens the final interpretation: under the current representation, adaptive safe steering is more important than local-vs-global structure alone. The key open problem for the next stage is therefore not whether latent steering works at all, but how to make it stronger without letting the safety constraints force the edit back to nearly zero.

A. Proofs and formal remarks

Proof of Proposition 1. Consider the local direction

$$d_{\mathrm{loc}}(z) = \mathrm{norm}\left(\frac{\sum_{j\in\mathcal{N}_k(z)} w_j(z)\,\delta_j}{\sum_{j\in\mathcal{N}_k(z)} w_j(z)}\right).$$

If $\mathcal{N}_k(z)$ is replaced by the entire bank and $w_j(z) \equiv c > 0$ is constant, then

$$d_{\mathrm{loc}}(z) = \mathrm{norm}\left(\frac{1}{m}\sum_{j=1}^{m}\delta_j\right),$$

which is exactly the normalized global average direction.

With constant weights, the denominator is $\sum_{j=1}^{m} c = mc$ and the numerator is $c\sum_{j=1}^{m}\delta_j$. Their ratio is

$$\frac{c\sum_{j=1}^{m}\delta_j}{mc} = \frac{1}{m}\sum_{j=1}^{m}\delta_j.$$

Normalizing this vector gives the normalized global average direction.

Proof of Proposition 2. Let $\mathcal{G}$ be a finite candidate grid, let $\mathcal{F}(z) \subseteq \mathcal{G}$ be the feasible subset for a given sample z, and let

$$\alpha^\star(z) \in \arg\max_{\alpha\in\mathcal{F}(z)} J(z,\alpha).$$

By definition, $\alpha^\star(z)$ is a maximizer of $J(z,\alpha)$ over the feasible set $\mathcal{F}(z)$. Therefore $J(z,\alpha^\star(z))$ is at least as large as the objective value at any other feasible point $\bar{\alpha} \in \mathcal{F}(z)$, so

$$J(z,\alpha^\star(z)) \geq J(z,\bar{\alpha}).$$

Remark on Proposition 2. The proposition does not claim that the *empirical* safe policy must dominate every fixed policy under every *external* metric. It states only that, under the *same internal selection objective and feasible grid*, the per-sample optimizer is the pointwise optimum. Empirically, the week-8 results show that this internal optimization correlates well with external preservation metrics.

B. Additional numerical details

Table 6: **Week-8 safe policy vs. matched week-7 fixed local_k8 baseline.**

Method	Digit acc	Consist.	Gain	Drift
Week-7 fixed local_k8	0.9677	0.9856	2.6936	4.5867
Week-8 auto local_k8	0.9722	1.0000	2.9347	4.2603

Table 7: **Week-10 tuned-global probe accuracies on dSprites.**

Target	ID acc	OOD acc
Shape	0.7429	0.6906
Scale	0.4913	0.5281

Table 8: **Week-11 auto-step statistics on dSprites.**

Split	Family	τ_{img}	Mean $\alpha^\star$	Nonzero rate
ID	Global	7	0.7199 ± 0.1577	0.7622 ± 0.1006
ID	Local	7	0.5151 ± 0.1150	0.6729 ± 0.1082
OOD	Global	7	0.7105 ± 0.1625	0.7566 ± 0.1012
OOD	Local	7	0.5000 ± 0.1110	0.6615 ± 0.1048

Table 9: **Week-11 independent preservation summary on dSprites.** Lower is better for all metrics.

Split	Family	Step	Chamfer	Topology break
ID	Global	Auto	0.0275 ± 0.0030	0.0021 ± 0.0012
ID	Local	Auto	0.0253 ± 0.0035	0.0036 ± 0.0030
OOD	Global	Auto	0.0273 ± 0.0027	0.0008 ± 0.0008
OOD	Local	Auto	0.0248 ± 0.0033	0.0018 ± 0.0025
OOD	Local	Fixed	0.0490 ± 0.0119	0.0933 ± 0.0329

C. Artifact provenance and audit trail

All reported scalar values in the paper were transcribed from archived generated artifacts. The audit trail used for this version is:

- **Week 5.** Run `20260224_074746_week5`; source file: `results.csv`; rule: direct baseline metrics from the only archived week-5 run.
- **Week 6.** Run `20260228_180215_week6`; source file: `results_week6.csv`; rule: fixed-step values reported at $\alpha = 1.0$ and $\alpha = 2.0$ from the single archived sweep.
- **Week 7.** Run `20260315_105429_week7_local_dirs`; source files: `results_stage3_by_alpha.csv` and `results_stage3_best_under_budget.csv`; rule: matched fixed-step comparison at $\alpha = 1.0$, with the best-under-budget local family taken from the archived summary table.

- **Week 8.** Run `20260315_113349_week8_safe_alpha`; source files: `results_stage4_auto.csv` and `results_stage4_compare_vs_fixed_a1.csv`; rule: highest-objective archived auto policy, with the matched comparison taken against the generated fixed-$\alpha = 1.0$ baseline table.
- **Week 9.** Run `20260315_232052_week9_beta_vae`; source files: `beta_recon_summary.csv` and `beta_best_auto.csv`; rule: reconstruction summary and best safe policy reported separately for each $\beta \in \{1, 2, 4\}$.
- **Week 10 headline tuned-global result.** Run `20260319_214326_week10_dsprites_ood_only` inside `output_vae_tuned.zip`; source files: `tables/selected_comparison.csv`, `tables/auto_alpha_summary.csv`, `tables/fixed_alpha_sweep.csv`, and `tables/probe_eval.csv`; rule: among tuned-global reruns with the same protocol, select the run with the highest OOD auto-steering objective.
- **Week 11 matched dSprites aggregation.** Five full global reruns `20260401_204921`, `20260401_211446`, `20260401_214018`, `20260401_220548`, and `20260401_223111` from the tuned-global pipeline, together with five full local reruns `20260331_035529`, `20260331_042059`, `20260331_044627`, `20260331_051153`, and `20260331_053721` from the local-kNN pipeline; source files: each run's `tables/selected_comparison.csv`, `tables/fixed_alpha_sweep.csv`, and `tables/auto_alpha_summary.csv`; aggregation outputs: `mean_std_summary.csv`, `final_table_pretty.csv`, and `headline_rows.csv`; rule: compute matched mean $\pm$ std summaries across the five reruns for both global and local families on ID and OOD splits.
- **Week 11 independent contour-aware preservation analysis.** Source artifact: `week11_independent_preservation_outputs.zip`; source files: `tables/indep_preservation_mean_std.csv`, `tables/indep_preservation_pretty.csv`, and the per-run `indep_preservation_selected_summary.csv` files for the same five global and five local reruns; rule: compute contour Chamfer, orientation drift, centroid drift, and topology break directly from thresholded reconstruction–edit pairs after largest-component extraction, centroid alignment, and area normalization.

References

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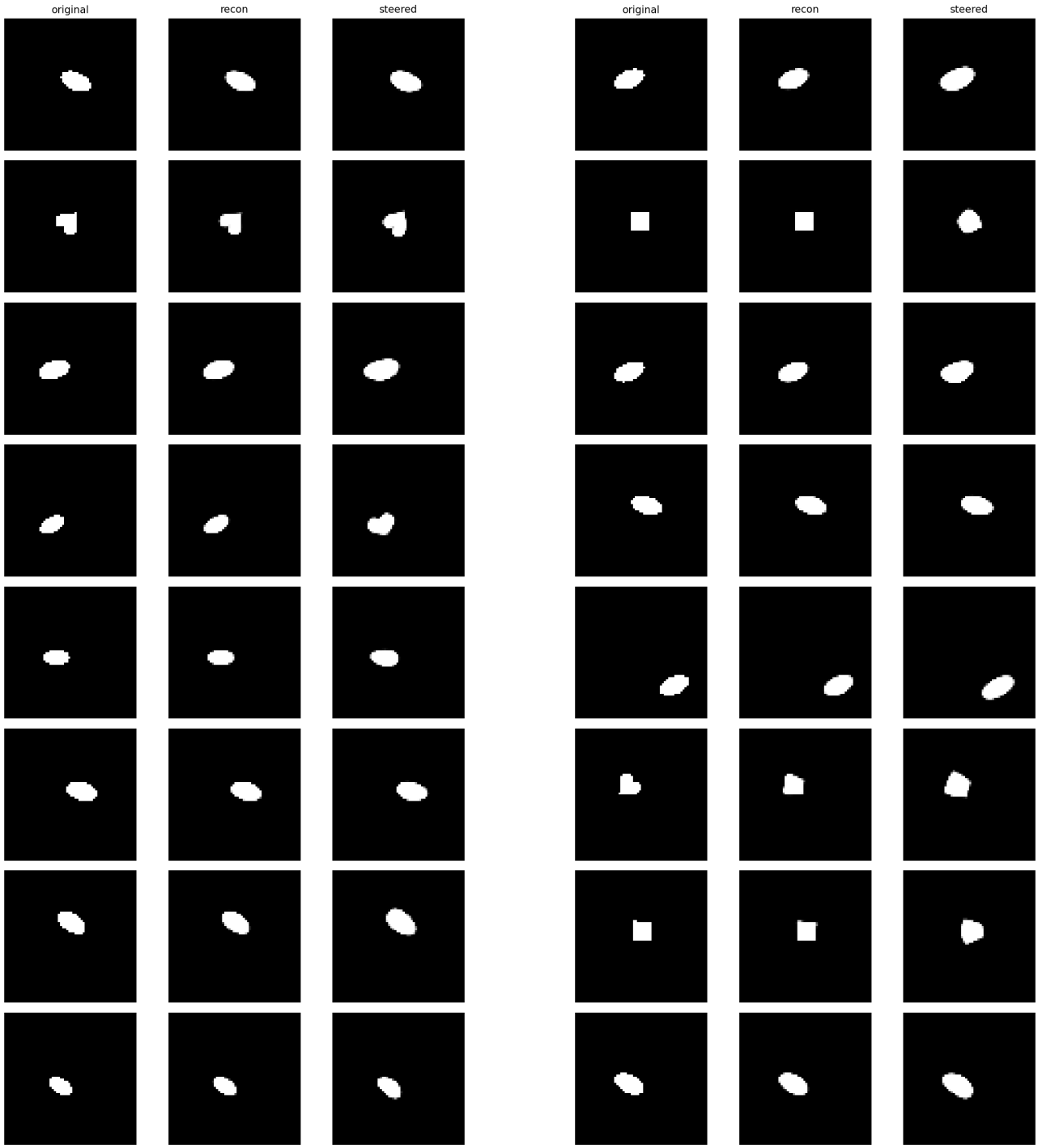

(a) ID preview (b) OOD preview

Figure 6: **Qualitative week-10 previews for tuned global dSprites steering.** The edit is visible but intentionally conservative in the safe regime; stronger fixed steps produce larger changes but noticeably worse preservation.